DATA SCIENCE TOOLS WORKSHOP

CDCSC19

PROJECT REPORT: SIGNBRIDGE – SIGN LANGUAGE

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Table of Contents

1 Case Studies/Research Papers Overview

1.1 Paper 1: Real-Time Turkish Sign Language Recognition using Mediapipe and YOLOv4 for Interactive Learning Systems

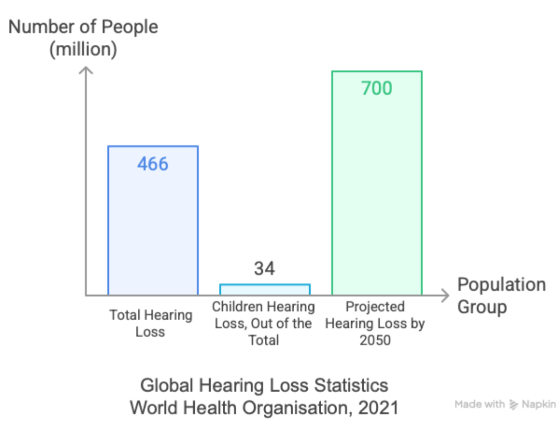
Authors: Melek Alaftekin, Ishak Pacal & Kenan Cicek.

Published in: Neural Computing and Applications, Springer, 15 Feb 2024

DOI: 10.1007/s00521-024-09503-6

1.1.1 Introduction

Sign language is a communication tool that deaf and dumb people use to convey their feelings, thoughts, desires etc. to the outside of their world. It has a nonverbal form of communication that uses hands, arm, head, facial gestures and body posture. According to the data of the world health organization in 2021, there are around 466 million people suffering from hearing loss around the world including 34 million children. It is estimated that this number will exceed 700 million by 2050 World Health Organization, 2021.



(a) WHO Data 2021

(b) Flow of System

Figure 1: System Flow Diagram

1.1.2 Methodology

Overall Approach In this section, the detailed structure of the proposed method is explained. In the first step, a dataset is created using a professional camera, outlining the approach followed by the suggested model presented visually in Fig. 1. Prior to commencing the model's training, a manual classification process is undertaken. As a result of this process, the emerged dataset consists of 1500 labelled images and is divided into random training, testing and validation stages at specified ratios. Images for each class are allocated as 70% for training, 15% for validation and 15% for testing. Data augmentation involves various techniques to alter the size and quality of training datasets. During training, augmentation techniques such as contrast adjustment, rotation, mirroring and brightness are utilized to increase data diversity. Due to the limited size of the Turkish sign language dataset for standalone training, different data augmentation techniques are applied to enhance the models' generalization ability on test data. With these two methods, both data size and diversity are increased.

Models In Use The YOLOv4 algorithm is employed as the object detection model due to its efficient optimization capabilities. YOLOv4 utilizes a three-head detector structure, with feature maps created at three different scales: 19x19, 38x38, and 76x76. The heads are responsible for detecting objects of varying sizes and outputting bounding box coordinates (x, y, width, height), confidence scores, and class predictions. After YOLOv4 identifies the hand and extracts the region of interest (ROI), the cropped image or the hand landmark data obtained through MediaPipe is passed to a Convolutional Neural Network (CNN) for gesture classification. The CNN is trained to recognize the hand gestures as corresponding to one of the 26 TSL alphabets.

Proposed Dataset The dataset used in this study is a custom collection of Turkish Sign Language (TSL) gestures, specifically static hand gestures representing the Turkish alphabet. The key features of the dataset are:

- **Type**: The dataset consists of static hand gestures corresponding to the Turkish alphabet (A–Z).
- **Capture Method**: Hand landmark data (21 key points per hand) is extracted using MediaPipe, which avoids full-frame image processing.
- **Source**: The dataset was recorded by the authors with gesture samples captured from a limited number of individuals.
- **Environment**: The data is collected in a controlled environment with uniform backgrounds and consistent lighting to reduce noise and improve recognition accuracy.
- **Format**: The data consists of (x, y, z) coordinates of the hand landmarks, which are used as feature vectors for training the CNN model.
- **Size**: Small but sufficient for real-time use (exact sample count unspecified, though 1,500 labelled images are mentioned earlier).

1.1.3 Results

Problem Addressed This paper addresses the problem of recognizing static Turkish Sign Language (TSL) gestures using real-time video input, aiming for a simple, efficient, and accurate recognition system suitable for resource-constrained environments.

Tools & Methods Used

- **MediaPipe**: Used to extract 21 hand landmarks from webcam input. Each hand pose is converted into normalized (x, y, z) coordinates.
- **CNN Model**: A lightweight Convolutional Neural Network is trained on this landmark data to classify each gesture into one of 26 English alphabets.
- **Dataset**: A custom dataset of Turkish Sign Language gestures recorded using a web-cam.
- **Training & Evaluation**: Achieved high accuracy (97%) on test data, with low latency suitable for real-time use.

Key Contributions

- Demonstrated that hand landmark data (instead of full-frame video) can be used effectively to classify TSL letters.
- The proposed system is efficient and can be deployed on resource-constrained devices such as web or mobile platforms.
- Despite the relatively small dataset, the CNN model achieves high accuracy, indicating the effectiveness of the approach.

Limitations Highlighted by Authors

- The system is limited to recognizing static gestures of the Turkish Sign Language alphabet and does not support dynamic gestures or sentence-level recognition.
- No user feedback, interactivity, or learning mechanisms integrated.
- Model performance may vary under different lighting or background conditions.
- Lacks diversity in dataset—only a small number of users and environments were used.

1.1.4 Comparative Discussion

Feature	Springer Paper	SignBridge (Proposed System)
Recognition Scope	Static TSL Alphabets	Alphabets, Words, and Sentences
Learning Mechanism	None	Gamified learning system
User Feedback	No feedback mechanisms	Real-time corrections, hints, scores
Tech Stack	MediaPipe + CNN	MediaPipe + TensorFlow + MERN stack
Deployment	Local, basic UI	Cross-platform, interactive frontend
Dataset Diversity	Low (single source)	Expandable, community-contributed

1.1.5 Research Gaps Identified

1. **Lack of Educational Layer:** The paper focuses solely on recognition but does not explore teaching or user engagement. SignBridge adds an educational dimension via levels, feedback, and gamification.
2. **Limited Scope of Recognition:** Shows restricted to static alphabet gestures. SignBridge expands this to words and simple sentences, encouraging natural progression in language learning.
3. **No Gamification or Motivation Strategy:** The paper lacks user engagement tools like achievements, scores, or progression maps. SignBridge introduces a game-like structure to motivate consistent learning.
4. **Absence of Real-Time Feedback**: Incorrect gestures aren't corrected in real time. SignBridge adds feedback loops to help users learn by doing.
5. **Scalability and Accessibility**: No mention of large-scale or web-based deployment. SignBridge is built on the MERN stack, making it web-deployable and scalable for classrooms and individual learners.

1.1.6 Conclusion

The reviewed Springer paper presents an efficient and real-time solution for static TSL alphabet recognition using MediaPipe and CNN. It lays a solid technical foundation by demonstrating that hand landmark data can be effectively used for gesture classification in real time with high accuracy. However, it remains limited to static alphabet recognition and lacks the educational and interactive features necessary for language acquisition. SignBridge builds upon these recognition techniques, transforming them into an engaging and gamified learning platform that promotes consistent learning through levels, real-time feedback, and web accessibility. This highlights a compelling research direction—shifting from gesture recognition to language learning via inclusive, scalable, and user-friendly platforms.

1.2 Paper 2: A Multi-lingual Sign Language Recognition System Using Machine Learning

Author: Fatma M. Najib
Published in: Multimedia Tools and Applications, Springer, 24 September 2024
DOI: 10.1007/s11042-024-20165-3

1.2.1 Introduction

This study falls under the first category which is sign language recognition. Many recent studies in the category of sign language recognition introduced and proposed systems for recognizing a specific sign language such as American Sign Language, Indian Sign Language, and Arabic Sign language. Each sign language has its own and different signs. In this paper, multiple languages recognition-based machine learning system is proposed which is called Multi-lingual Sign Languages Interpreter (MSLI) system. The machine learning (ML) model of the proposed MSLI can be learnt using a specific sign language or multiple combined sign language.

1.2.2 Methodology

Overall Approach The MSLI system is developed to recognize hand signs from multiple languages. The approach involves training machine learning models on datasets comprising various sign languages, enabling the system to detect the language of input signs and accurately interpret them. The system also incorporates a two-step recognition process for sequences of signs in the same language, enhancing efficiency by detecting the language once and applying it to subsequent signs.

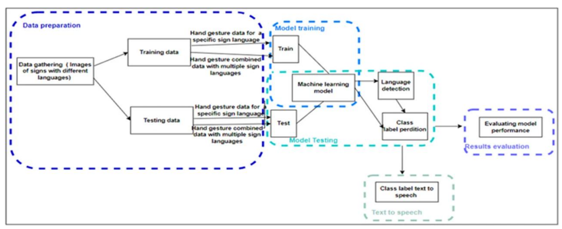


Figure 2: The architecture of proposed MSLI system

Models In Use This subsystem used the input hand gesture images to train the machine learning model. Any supervised machine learning algorithm(s) can be used. MSLI can separately build a machine learning model for each dataset with a specific sign language. Also, a machine learning model can be built for a combined dataset with different sign languages. MSLI is trained to detect the sign and the language of hand gesture images.

A machine learning model for a specific language is tested using hand gesture images of the same language. MSLI can predict the class label of signs. Also, hand gesture images with different languages can be tested using the machine learning model of the combined dataset. MSLI can predict the language and sign label of the testing images. In the case of sequence of signs, this subsystem can follow two steps recognition mode. Firstly, the language is detected for the first sign. Then, the rest signs are classified according to the detected sign language.

Proposed Dataset The study utilizes 11 datasets encompassing various sign languages, including American Sign Language (ASL), Indian Sign Language (ISL), and Arabic Sign Language (ArSL). These datasets provide a diverse range of hand signs, facilitating the training of models capable of recognizing multiple sign languages. Data is divided into training and testing data.

1.2.3 Results

Problem Addressed The research addresses the challenge of developing a unified system capable of recognizing multiple sign languages, thereby overcoming the limitations of language-specific recognition systems. We generate a multi-lingual combined dataset from the 11 datasets with different languages. A class label of our combined dataset is a combination of sign name and language name. A ML.NET model was built for the combined dataset of 11 languages. Classification performance was measured.

Tools & Methods Used

- Machine learning models trained on individual and combined sign language datasets.
- Language detection mechanisms to identify the language of input signs.
- Two-step recognition process for sequences of signs in the same language.

Key Contributions

- Development of the MSLI system capable of recognizing multiple sign languages.
- Implementation of language detection to enhance recognition accuracy.
- Demonstration of high accuracy rates in both separate and combined language settings.

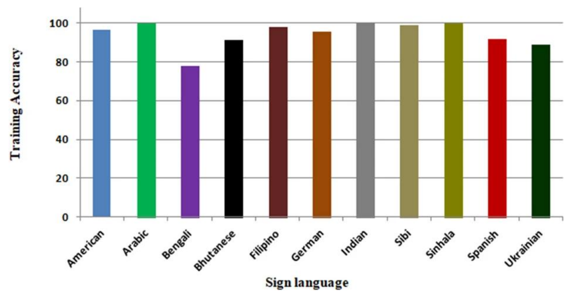


Figure 3: Training accuracy percentage of each separate sign language dataset

1.2.4 Comparative Discussion

Feature	MSLI System (Najib, 2024)	SignBridge (Proposed System)
Recognition Scope	Multiple sign languages	Alphabets, words, sentences
Learning Mechanism	Machine learning models	Gamified learning system
User Feedback	Not specified	Real-time corrections, hints
Tech Stack	Machine learning	MediaPipe, TensorFlow, MERN
Deployment	Not specified	Cross-platform, interactive
Dataset Diversity	11 datasets	Expandable, community-driven

1.2.5 Research Gaps Identified

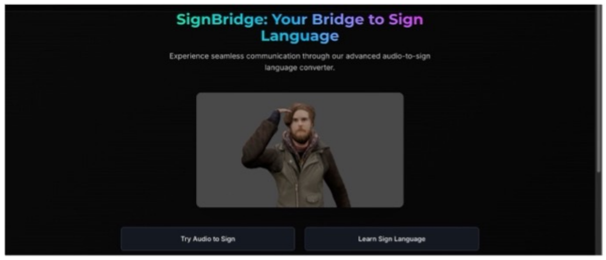
1. **Educational Layer:** The MSLI system focuses on recognition without incorporating teaching or user engagement mechanisms.
2. **Scope of Recognition:** While MSLI handles multiple sign languages, it does not explicitly address the recognition of words or sentences.
3. **User Engagement:** The system lacks gamification elements that could enhance user motivation and learning.
4. **Real-Time Feedback:** Immediate feedback mechanisms for incorrect gestures are not discussed.
5. **Scalability & Accessibility:** Details regarding large-scale deployment or web-based accessibility are not provided.

1.2.6 Conclusion

The MSLI system represents a significant advancement in sign language recognition, offering a unified approach to interpreting multiple sign languages using machine learning. Its high accuracy rates in both separate and combined language settings underscore its potential for real-world applications. However, the system's focus remains on recognition, with limited emphasis on educational features, user engagement, and real-time feedback. Future developments could explore these aspects to create more comprehensive and user-friendly sign language learning platforms.

2 Project Report: SignBridge – A Real-Time Sign Language Learning and Detection Platform

2.1 Abstract



This project presents **SignBridge**, an interactive sign language learning and recognition platform that leverages deep learning and pose estimation to recognize static, dynamic, and sentence-level American Sign Language (ASL) signs. The system integrates a hybrid CNN-LSTM model for multi-modal detection and a web-based game interface for users to learn, practice, and test their skills. Key innovations include gamification of the SLR concept, focusing on user engagement on multi-level sign recognition using real-time webcam input and text-to-sign video generation, making sign language more accessible and engaging for diverse users.

2.2 Introduction

Sign languages serve as a primary means of communication for the deaf and hard-of-hearing communities. However, due to limited exposure and educational resources, many people remain unfamiliar with sign language. Bridging this gap requires accessible, interactive, and intelligent learning systems.

This project aims to develop a real-time sign language recognition model capable of detecting alphabets (static signs), words (dynamic gestures), and sentence-level communication. Integrated into a gamified web platform, the system fosters active learning, self-assessment, and educational support.

2.3 Dataset Overview

To build a comprehensive model capable of understanding different complexities of sign language, we used:

- **ASL Alphabet Dataset**
  - Type: Static image dataset
  - Content: 26 English alphabets (A-Z) in ASL
  - Application: Level 1 - Alphabet detection
- **WLASL Processed Dataset**
  - Type: Video dataset
  - Content: Over 2,000 sign language words with diverse signers
  - Application: Level 2 & 3 - Word and sentence-level detection
- **Custom Datasets by Team Members**: Curated multiple videos from different angles and viewpoints for words and simple sentences.
- **Keypoint-Based Datasets**
  - Feature extraction using **MediaPipe** for hand, face, and pose landmarks
  - Used to reduce input complexity and enhance robustness against background noise

2.4 Technology Stack

2.4.1 Backend & AI Stack

- **TensorFlow**, **Keras** – For model building and training
- **MediaPipe** – Pose estimation and keypoint extraction
- **OpenCV**, **NumPy**, **Matplotlib** – Preprocessing, visualization, and input handling
- **MongoDB** – Storage for user progress and data

2.4.2 Frontend & Web Integration

- **React.js**, **Tailwind CSS** – UI/UX and interactive frontend design
- **JavaScript** – Logic, event handling, webcam integration
- **app.py (Flask)** – Backend API and model integration
- **Webcam API** – For capturing and classifying user gestures in real-time

2.5 Model Architecture and Performance

```
keypoints = extract_hand_keypoints(results)
print("Keypoints shape:", keypoints.shape)
print("Keypoints array:", keypoints)
```

```
Keypoints shape: (63,)
Keypoints array: [ 5.76632380e-01  7.34734297e-01 -1.99082297e-06  4.08674240e-01
  6.79440260e-01 -5.49425669e-02  2.57385612e-01  5.62679350e-01
 -8.68138224e-02  2.04333678e-01  4.11582232e-01 -1.21630989e-01
  2.75090575e-01  3.25372219e-01 -1.41214088e-01  3.56961071e-01
  3.60871851e-01 -1.20472955e-02  3.59862566e-01  2.38251179e-01
 -9.96847749e-02  3.69555265e-01  3.79254937e-01 -1.41396388e-01
  3.74710381e-01  4.54117388e-01 -1.58999190e-01  4.67347324e-01
  3.52833062e-01 -2.31628735e-02  4.52521920e-01  2.39827722e-01
 -1.18539929e-01  4.60523158e-01  4.09561753e-01 -1.33148760e-01
  4.67289686e-01  4.64227825e-01 -1.21915884e-01  5.75082719e-01
  3.62058222e-01 -4.92131673e-02  5.62810063e-01  2.46496603e-01
 -1.43304959e-01  5.51627636e-01  4.13325012e-01 -1.05348952e-01
  5.48109055e-01  4.74757373e-01 -5.53499833e-02  6.83713913e-01
  3.90895158e-01 -8.06467384e-02  6.61004663e-01  2.99467862e-01
 -1.29065439e-01  6.37284219e-01  4.17855978e-01 -9.21027511e-02
  6.29212976e-01  4.70569015e-01 -5.12687676e-02]
```

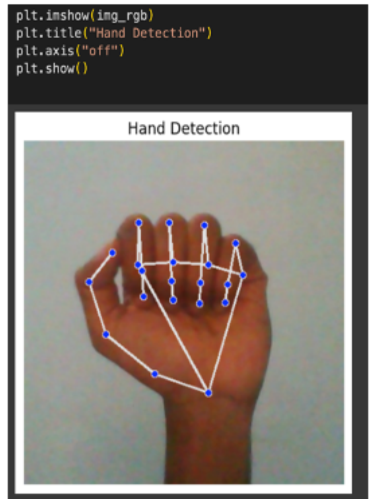
The sign language recognition component of SignBridge is divided into two primary subsystems: static gesture recognition for alphabets and dynamic gesture recognition for words and sentences. Both systems leverage keypoint-based input features extracted via MediaPipe, which efficiently isolates hand landmarks for consistent, background-invariant input representation.

2.5.1 Static Gesture Recognition

This subsystem focuses on recognizing individual ASL alphabets using images from a webcam or dataset.

Preprocessing and Feature Extraction

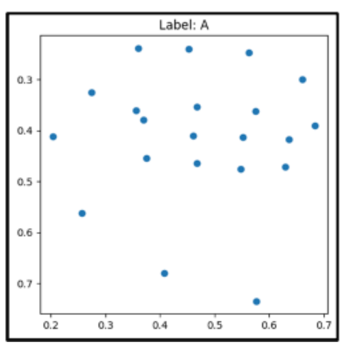
- **Dataset**: ASL Alphabet Dataset (A–Z) from Kaggle, WLASL dataset
- **Hand Keypoint Extraction**: MediaPipe Hands module extracts 21 hand landmarks (x, y, z), producing a 63-dimensional feature vector per frame.
- **Normalization**: Input keypoints are normalized to improve model generalization and reduce inter-user variability.




```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 lstm (LSTM)                 (None, 64)                32768

 dropout (Dropout)           (None, 64)                0

 dense (Dense)               (None, 64)                4160

 dense_1 (Dense)             (None, 28)                1820

=================================================================
Total params: 38,748
Trainable params: 38,748
Non-trainable params: 0
_________________________________________________________________
```

Model Architecture A lightweight sequential model was designed using TensorFlow and Keras to optimize for real-time inference:

- Input Shape: (1, 63) – mimicking a single time-step for LSTM input
- Output: Softmax layer with 26 neurons representing ASL alphabets
- Loss Function: Categorical Crossentropy
- Optimizer: Adam

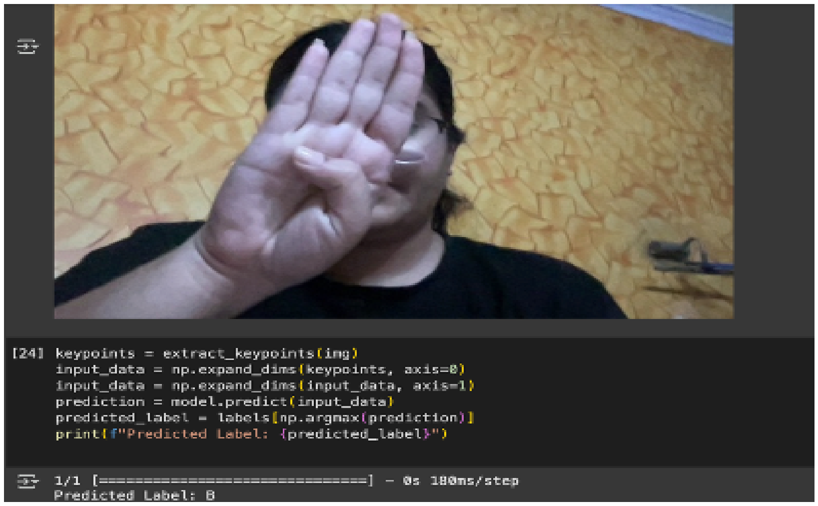


Training & Evaluation

- **Epochs**: 15
- **Batch Size**: 32
- **Train-Test Split**: 80:20 using train_test_split
- **Label Encoding**: Done via LabelEncoder and to_categorical

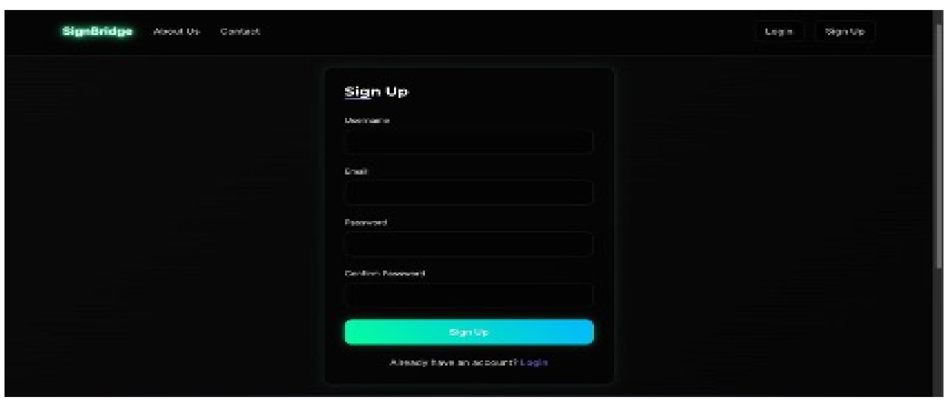
Performance:

- **Test Accuracy: 95% for sentences, 100% in case of alphabets and words**



Inference Pipeline

- A live webcam stream is processed in real-time using OpenCV.
- Extracted hand keypoints are passed to the trained LSTM model.
- Predicted labels are overlaid on the video stream for interactive feedback.



2.6 Dynamic Model Architecture and Performance

This system is designed to recognize dynamic American Sign Language (ASL) gestures—primarily words and short phrases—from videos or live webcam input. It combines MediaPipe Holistic for spatiotemporal feature extraction with a deep learning model built using LSTM (Long Short-Term Memory) networks to capture the temporal dynamics of gestures.

Structure:

- Each sample is a video representing a single ASL word or phrase.
- Labels are extracted from the filename (custom dataset) or the gloss field in the WLASL metadata.

2.6.1 Feature Extraction

MediaPipe Holistic is used to extract body, hand, and facial landmarks for each frame in a video.

Extracted Keypoints:

- **Pose Landmarks**: 33
- **Face Landmarks**: 468
- **Left Hand Landmarks**: 21
- **Right Hand Landmarks**: 21
- **Total**: 543 landmarks per frame

Sequence Processing:

- Each video is processed to extract up to **30 frames**.
- If fewer than 30 frames are available, zero-padding is applied.
- The resulting input shape per video is **(30, 1629)**.

2.6.2 Data Preprocessing

- All keypoint sequences are standardized to a uniform shape using padding/truncation: **(30, 1629)**
- Labels are encoded using LabelEncoder and converted to one-hot vectors using to_categorical().
- The dataset is saved as serialized arrays for efficient loading during training.

2.6.3 Model Architecture

A sequential neural network model using LSTM layers is used to capture temporal dependencies between video frames.

Architecture Overview:

- **Input Shape**: (30, 1662) (Note: 1662 includes additional hand keypoints in some versions of the pipeline)
- **Layers**:
  - Masking layer to handle zero-padded time steps
  - LSTM(128) with return_sequences=True
  - Dropout(0.3)
  - LSTM(64)
  - Dense(num_classes) with softmax activation

Compilation:

- **Loss Function**: Categorical Crossentropy
- **Optimizer**: Adam
- **Evaluation Metric**: Accuracy.

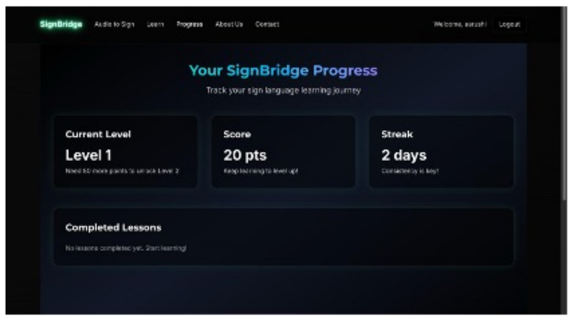
2.6.4 Real-Time Inference

A webcam-based real-time inference module is integrated to demonstrate live ASL gesture recognition.

Procedure:

- Capture frames from the webcam.
- Extract keypoints from each frame using MediaPipe Holistic.
- Maintain a buffer of the most recent 31 frames.
- When the buffer is full, the sequence is passed to the trained model for classification.
- The predicted label and confidence score are displayed in real time on the video feed.

2.7 Platform Overview: SignBridge – A Gamified Sign Language Learning Experience



SignBridge is an interactive web-based learning platform designed to make American Sign Language (ASL) education accessible, engaging, and effective. Combining machine learning and gamification, the platform supports real-time sign detection through webcam input and offers a structured, user-friendly curriculum across three progressive levels:

- **Level 1: Alphabets**
- **Level 2: Words**
- **Level 3: Sentences**

2.7.1 Key Features

- **Real-Time Sign Feedback**: Webcam-based ASL detection allows users to receive instant correction and validation of signs during practice sessions.
- **Text/Audio-to-Sign Conversion**: Users can input text or speak into their device, and the system will return the equivalent ASL gesture video or animation. NLTK model being used
- **Gamified Progress Tracking**:
  - Points awarded for correct answers
  - Streak bonuses
  - Visual indicators of completed lessons
  - Performance reports to monitor growth and unlock new levels
- **Learn & Practice Modes**:
  - **Learn Mode**: Introduces signs through videos and visual breakdowns.
  - **Practice Mode**: Enables webcam-based gesture input and provides feedback on accuracy and timing.
  - **Progressive Curriculum**: Encourages mastery through sequential content unlocking and increasing complexity.

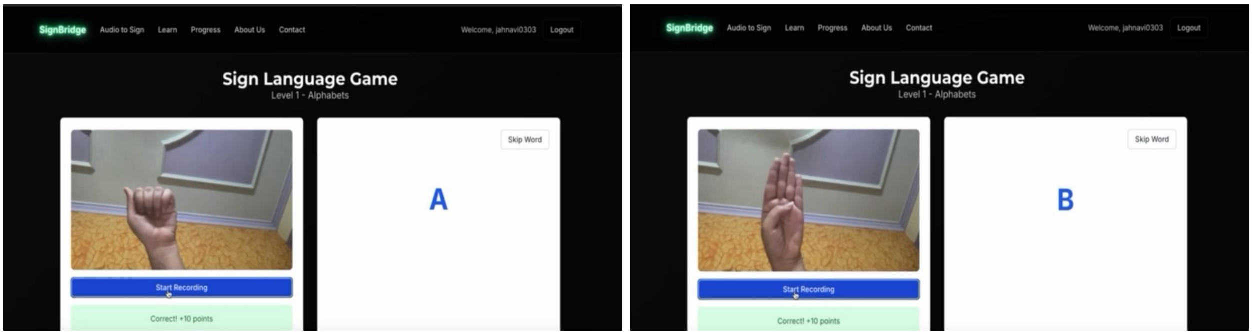


Figure 4: Real-time feedback by sharing incorrect when wrong sign is made

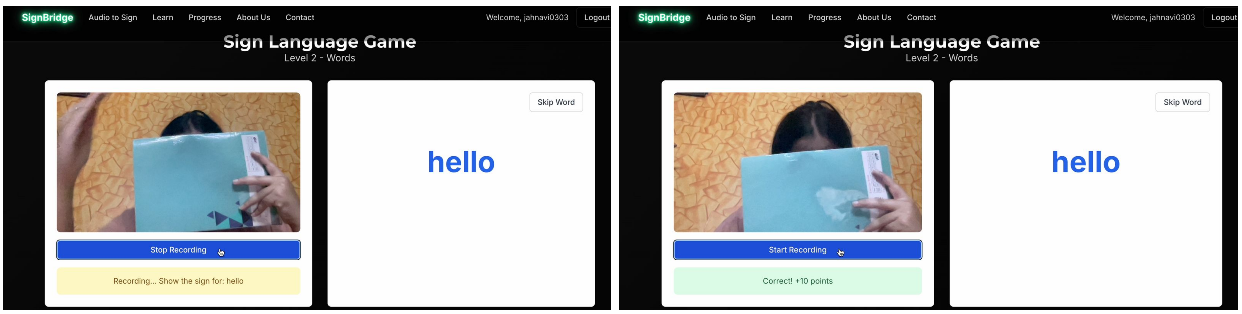


Figure 5: Points awarded for correct recognition of words and simple sentences

2.7.2 Model Integration

Both models (alphabet and word/sentence recognition) are exported using the .h5 format and loaded into the backend using Flask (app.py). Inference is performed upon webcam input, and predictions are routed to the front-end interface via real-time communication.

The system supports:

- **On-the-fly predictions**
- **Instant feedback rendering**

2.8 Use Cases

The SignBridge platform and its underlying sign language recognition system present several impactful applications across education, accessibility, and professional training domains:

2.8.1 Inclusive Education Platforms

The integration of real-time sign language recognition supports accessible and inclusive learning environments in schools, universities, and e-learning platforms. It empowers deaf and hard-of-hearing students to interact with content in their native visual language while allowing hearing students to develop sign language literacy, fostering mutual communication and empathy.

2.8.2 Public Sector and Frontline Workforce Training

SignBridge can be deployed in training programs for public service professionals, including law enforcement, healthcare providers, transportation staff, and emergency responders. By learning essential sign language through interactive and gamified modules, staff can offer better support and communication in critical, real-world scenarios involving individuals with hearing or speech impairments.

2.8.3 Assistive Communication Technologies

As part of broader assistive technology suites, the platform serves as a communication aid for non-verbal or speech-impaired users. The ability to recognize signs in real time and convert spoken/written language into sign video outputs offers bi-directional interaction between users and their environments, enhancing autonomy and quality of life.

2.8.4 Digital Language Learning Platforms

The gamified structure of SignBridge aligns well with language learning and ed-tech applications, where ASL or other sign languages are taught interactively. The real-time feedback mechanism enhances engagement and accelerates skill acquisition, making it a valuable add-on for platforms like Duolingo, Coursera, or Khan Academy.

2.8.5 Workplace Accessibility Integration

Organizations striving for inclusive workplaces can embed *SignBridge* as part of accessibility compliance programs, enabling smoother interaction with deaf employees and customers. It supports internal communication training, onboarding, and workplace cultural sensitization initiatives.

2.9 Discussion

The integration of spatial (CNN) and temporal (LSTM) models effectively supports recognition across varying complexities in sign language—from static hand postures to dynamic gesture transitions. The use of MediaPipe for keypoint-based extraction reduces sensitivity to environmental noise, enhances speed, and supports real-time deployment.

By embedding the recognition system into an interactive web game, SignBridge overcomes the monotony of conventional learning tools, increases engagement, and lowers the barrier to entry for new learners. Furthermore, the modular design ensures that new datasets, sign languages, or features (e.g., grammar-aware modeling) can be added with minimal structural modification.

2.10 Conclusion

SignBridge provides a comprehensive learning experience, combining deep learning-based gesture recognition with gamified educational elements. The project highlights the potential of combining advanced AI models with user-centred design to create impactful assistive technologies.

2.10.1 Research Contributions and Gaps Addressed

This project, **SignBridge**, strategically addresses the above gaps through the following contributions:

- **Gamified Learning Architecture**: Incorporating levels (alphabets → words → sentences), score-based feedback, and progression logic transforms sign language learning into an interactive and motivating experience.
- **Text-to-Audio Conversion**: User gets the functionality to record their audios and see the visual ASL depiction for their input statement, on our interface.
- **Progress Record**: The user can see how much progress they have made in the form of levels cleared, points obtained, and no. of continuous streak days, helping them track their improvements over time.
- **Multi-Tier Deep Learning Pipeline**: A hybrid CNN-LSTM model enables robust detection of both **static gestures** and **dynamic sequences**, supporting multi-granular recognition from single letters to continuous sentence-level phrases.
- **Practical Deployment via Web Platform**: By integrating TensorFlow, MediaPipe, and React-based frontend with backend support via MongoDB and Flask (app.py), the system ensures real-world accessibility and usability.

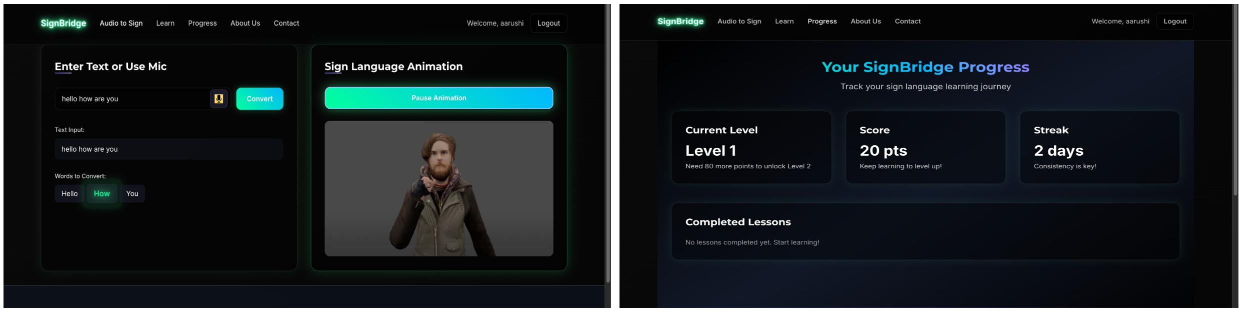


Figure 6: Research Gaps Solved Through SignBridge's Gamification

2.11 Future Work

- Extend the system to support other sign languages (e.g., Indian Sign Language, British Sign Language)
- Incorporate grammar-aware models and contextual translation for syntactic accuracy
- Develop cross-platform mobile applications using TensorFlow Lite or ONNX
- Explore personalized learning paths using user analytics and adaptive feedback mechanisms.

References:

1. **ASL Alphabet Dataset**
   Source: Kaggle – American Sign Language
2. **WLASL Processed Dataset**
   Source: Kaggle – WLASL Processed
3. **Multi-lingual sign language recognition system using machine learning:** `https://link.springer.com/article/10.1007/s11042-024-20165-3`
4. **Real-time sign language recognition based on YOLO algorithm:** `https://link.springer.com/article/10.1007/s00521-024-09503-6`